

HW0324

```
library(readxl)
```

```
## Warning: 套件 'readxl' 是用 R 版本 4.2.3 來建造的
```

```
collegetown <- read_excel("C:/finance ecomatrics/data/collegetown.xlsx")
View(collegetown)
```

a

```
mod_a <- lm(log(price) ~ sqft, data = collegetown)
summary(mod_a)
```

```
##
## Call:
## lm(formula = log(price) ~ sqft, data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.32528 -0.19199  0.00122  0.21678  0.86609
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.393866   0.043288  101.50  <2e-16 ***
## sqft         0.036044   0.001486   24.26  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.34 on 498 degrees of freedom
## Multiple R-squared:  0.5417, Adjusted R-squared:  0.5408
## F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
mean_price <- mean(collegetown$price)
mean_sqft <- mean(collegetown$sqft)
b2_a <- coef(mod_a)[[2]]
slope_a <- b2_a * mean_price
elas_a <- b2_a * mean_sqft
cat("Slope at means (a) =", slope_a, "\n")
```

```
## Slope at means (a) = 9.019661
```

```
cat("Elasticity at means (a) =", elas_a, "\n\n")
```

```
## Elasticity at means (a) = 0.9833701
```

$$\ln(PRICE) = 4.393866 + 0.036044 * SQFT + e$$

1. Slope Coefficient ($\hat{\beta}_2 = 0.0360$): holding all else constant, a one-unit increase in house size (*sqft*) is associated with an expected 3.6% increase in the house price.
2. Slope at the Means (9.02) : This indicates that for a house of average value, an additional unit of living area increases the expected price by approximately 9.02 units of currency
3. Elasticity at the Means (0.98) : 1% increase in house size is expected to lead to approximately a 0.983% increase in the house price

b

```
mod_b <- lm(log(price) ~ log(sqft), data = collegetown)
summary(mod_b)
```

```
##
## Call:
## lm(formula = log(price) ~ log(sqft), data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.36864 -0.20849  0.00487  0.23204  1.21099
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.04965     0.15797   12.97  <2e-16 ***
## log(sqft)    1.02483     0.04839   21.18  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3643 on 498 degrees of freedom
## Multiple R-squared:  0.4738, Adjusted R-squared:  0.4728
## F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
a2_b <- coef(mod_b)[[2]]
slope_b <- a2_b * (mean_price / mean_sqft)
elas_b <- a2_b

cat("Slope at means (b) =", slope_b, "\n")
```

```
## Slope at means (b) = 9.399923
```

```
cat("Elasticity at means (b) =", elas_b, "\n\n")
```

```
## Elasticity at means (b) = 1.024828
```

$$\ln(PRICE) = 2.04965 + 1.02483 * \ln(SQFT) + e$$

1. Elasticity Coefficient ($\hat{\alpha}_2 = 1.0248$): holding all else constant, a 1% increase in house size (*sqft*) is associated with an expected 1.025% increase in the house price.
2. Elasticity at the Means (1.024828): the elasticity is constant. Regardless of the house size, a 1% increase in square footage will always lead to a roughly 1.025% increase in price.

3. Slope at the Means (9.40): This indicates that for an “average” house, an additional 1-unit increase in area (1 sqft) raises the expected absolute price by approximately 9.40 units

C

```
mod_c <- lm(price ~ sqft, data = collegetown)
r2_linear <- summary(mod_c)$r.squared

pred_price_a <- exp(fitted(mod_a))
gen_r2_a <- cor(collegetown$price, pred_price_a)^2

pred_price_b <- exp(fitted(mod_b))
gen_r2_b <- cor(collegetown$price, pred_price_b)^2

cat("Linear Model R^2 =", r2_linear, "\n")
```

```
## Linear Model R^2 = 0.6413167
```

```
cat("Log-Linear Gen R^2 =", gen_r2_a, "\n")
```

```
## Log-Linear Gen R^2 = 0.6621612
```

```
cat("Log-Log Gen R^2 =", gen_r2_b, "\n\n")
```

```
## Log-Log Gen R^2 = 0.6445084
```

The results indicate that the Log-Linear model yields the highest Generalized R^2 (0.662). This signifies that, when evaluated on the original price scale, the log-linear specification explains the largest proportion of the variance in actual house prices. Therefore, based on the goodness of fit, the log-linear model outperforms the other functional forms.

d

```
library(tseries)
```

```
## Warning: 套件 'tseries' 是用 R 版本 4.2.3 來建造的
```

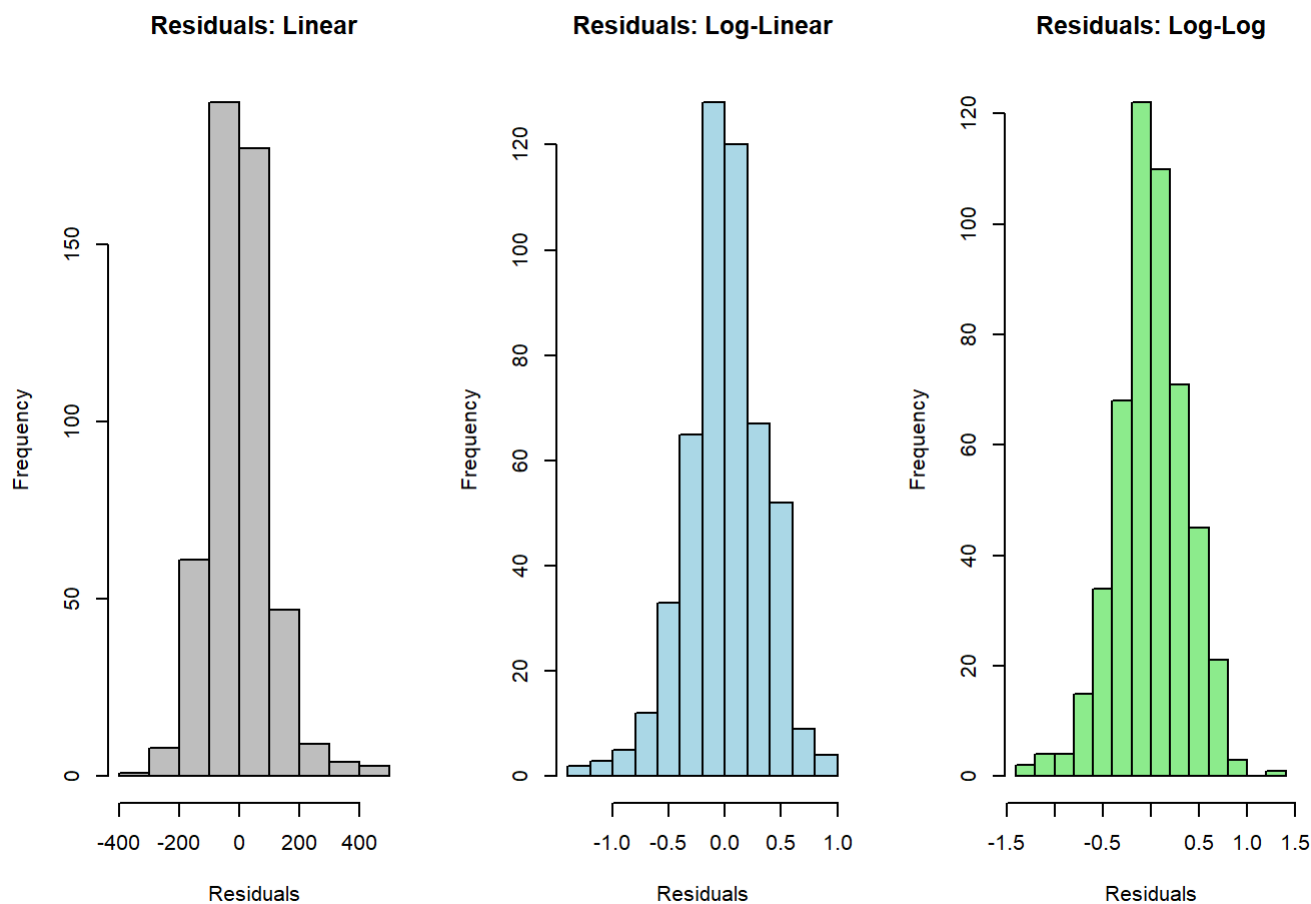
```
## Registered S3 method overwritten by 'quantmod':
##   method              from
##   as.zoo.data.frame zoo
```

```

res_a <- resid(mod_a)
res_b <- resid(mod_b)
res_c <- resid(mod_c)

# histogram
par(mfrow = c(1, 3))
hist(res_c, main = "Residuals: Linear", xlab = "Residuals", col = "gray")
hist(res_a, main = "Residuals: Log-Linear", xlab = "Residuals", col = "lightblue")
hist(res_b, main = "Residuals: Log-Log", xlab = "Residuals", col = "lightgreen")

```



```

# Jarque-Bera 常態檢定
jb_a <- jarque.bera.test(res_a)
jb_b <- jarque.bera.test(res_b)
jb_c <- jarque.bera.test(res_c)

cat("JB Test p-value (Linear) =", jb_c$p.value, "\n")

```

```
## JB Test p-value (Linear) = 0
```

```
cat("JB Test p-value (Log-Linear) =", jb_a$p.value, "\n")
```

```
## JB Test p-value (Log-Linear) = 1.609651e-06
```

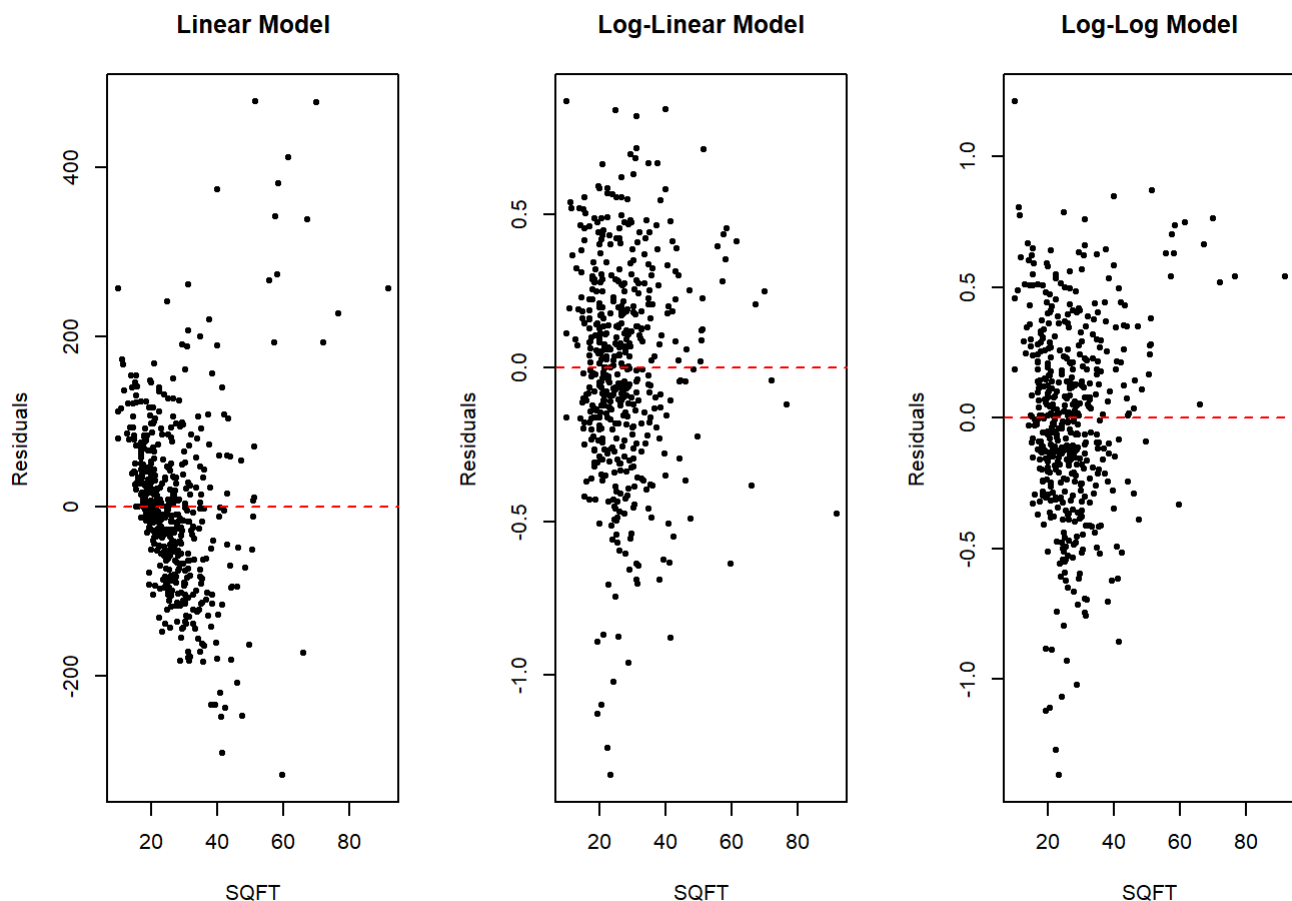
```
cat("JB Test p-value (Log-Log) =", jb_b$p.value, "\n")
```

```
## JB Test p-value (Log-Log) = 0.0007932632
```

If the p-value is less than 0.05, we reject the assumption of normality. Based on the result, these 3 models are not normally distributed.

e

```
par(mfrow = c(1, 3))
plot(collegetown$sqft, res_c, main = "Linear Model", ylab = "Residuals", xlab = "SQFT", pch=20)
abline(h = 0, col = "red", lty = 2)
plot(collegetown$sqft, res_a, main = "Log-Linear Model", ylab = "Residuals", xlab = "SQFT", pch=20)
abline(h = 0, col = "red", lty = 2)
plot(collegetown$sqft, res_b, main = "Log-Log Model", ylab = "Residuals", xlab = "SQFT", pch=20)
abline(h = 0, col = "red", lty = 2)
```



```
par(mfrow = c(1, 1))
```

Linear Model: We can clearly observe a “funnel shape” (or cone shape) pattern in the scatter plot. For smaller houses (*SQFT* between 20 and 40), the residuals are tightly clustered near the zero line. However, as the house size increases, the spread (variance) of the residuals widens significantly. This indicates the presence of severe heteroskedasticity. It means the linear model’s predictions become increasingly unreliable for larger properties.

Log Models (Middle and Right Plots): After applying a natural logarithm transformation to the dependent variable ($\ln(PRICE)$), the funnel shape present in the linear model is eliminated. In both the log-linear and log-log plots, the residuals are scattered much more uniformly around the zero line across all levels of $SQFT$. The spread no longer systematically increases with house size.

Overall Conclusion: This visually demonstrates that the log transformation is a highly effective technique. It successfully compresses the extreme variances associated with higher-priced homes, stabilizes the error variance, and corrects the heteroskedasticity issue. Consequently, the log-transformed models better satisfy the classical OLS assumptions.

f&g

```
new_data <- data.frame(sqft = 27)
```

```
# 1. Linear Model
```

```
pred_c <- predict(mod_c, newdata = new_data, interval = "prediction", level = 0.95)
cat("--- Linear Model ---\n")
```

```
## --- Linear Model ---
```

```
print(pred_c)
```

```
##          fit      lwr      upr
## 1 246.4557 44.27727 448.6341
```

```
# 2. Log-Linear Model (預測出  $\ln(PRICE)$ ，需取指數轉回)
```

```
# 取預測區間
```

```
pred_log_a <- predict(mod_a, newdata = new_data, interval = "prediction", level = 0.95)
```

```
# 計算修正後的點預測期望值:  $\exp(\text{fit} + \text{sigma}^2 / 2)$ 
```

```
sigma2_a <- summary(mod_a)$sigma^2
```

```
point_pred_a <- exp(pred_log_a[1, "fit"] + sigma2_a / 2)
```

```
# 區間上下界直接取指數
```

```
ci_lwr_a <- exp(pred_log_a[1, "lwr"])
```

```
ci_upr_a <- exp(pred_log_a[1, "upr"])
```

```
cat("\n--- Log-Linear Model ---\n")
```

```
##
```

```
## --- Log-Linear Model ---
```

```
cat("Point Prediction:", point_pred_a, "\n")
```

```
## Point Prediction: 226.9817
```

```
cat("95% PI: [", ci_lwr_a, ",", ci_upr_a, "]\n")
```

```
## 95% PI: [ 109.7685 , 418.1167 ]
```

```

pred_log_b <- predict(mod_b, newdata = new_data, interval = "prediction", level = 0.95)
sigma2_b <- summary(mod_b)$sigma^2
point_pred_b <- exp(pred_log_b[1, "fit"] + sigma2_b / 2)
ci_lwr_b <- exp(pred_log_b[1, "lwr"])
ci_upr_b <- exp(pred_log_b[1, "upr"])
cat("\n--- Log-Log Model ---\n")

```

```

##
## --- Log-Log Model ---

```

```

cat("Point Prediction:", point_pred_b, "\n")

```

```

## Point Prediction: 243.1514

```

```

cat("95% PI: [", ci_lwr_b, ",", ci_upr_b, "]\n")

```

```

## 95% PI: [ 111.1406 , 465.8406 ]

```

h

Based on the comprehensive analysis in this problem, the Log-Linear Model slightly outperforms the others.

1. Correcting Heteroskedasticity (Evidence from Part e): The log transformations effectively compressed the variance of higher-priced homes, stabilizing the error term and satisfying the crucial OLS assumption of homoskedasticity.

2. Superior Goodness-of-Fit (Evidence from Part c): By calculating the Generalized R^2 to evaluate the models on the original price scale fairly, the Log-Linear model achieved the highest proportion of explained variance ($Gen R^2 = 0.662$), outperforming both the linear and log-log specifications.